

Test Plan

1. **Objective** The primary goal is to verify the functional stability, API reliability, and baseline performance of the Klodr ERP module while staying within a **safe testing scope**.

2. Scope of Testing

- **In-Scope:** Functional UI testing, API contract validation, low-load performance testing, and "safe" security checks (Auth, RBAC).
- **Out-of-Scope:** Stress testing, intrusive penetration testing, and testing in the Production environment.

3. Methodology & Techniques

Klodr SaaS platform was driven by five distinct black-box techniques to ensure maximum coverage with structured logic.

- **Equivalence Partitioning (EP):** Used to divide input data into valid and invalid classes. For example, password testing was partitioned into classes like "Letters only" (Invalid) and "Alphanumeric" (Valid).
- **Boundary Value Analysis (BVA):** Focused on testing the "edges" of input ranges. You specifically tested the Klodr password length limit (range 12–32 characters), identifying a defect where the 33rd character was accepted.
- **Decision Table Testing:** Represented complex business logic in tabular format. This was applied to **File Upload** logic, where a file is only accepted if both the **Type** (.pdf/.jpg) and **Size** ($\leq 5\text{MB}$) conditions are met.
- **State Transition Testing:** Verified how the system moves between operational phases. For example, an invoice transitioning from "Active/Paid" to "Refunded/Credit Note Issued" upon clicking the **Credit Note** button.
- **Exploratory Testing:** Leveraged your intuition and risk-based thinking to find unexpected bugs. A key discovery here was the **missing "Delete" functionality** for erroneous appointment records.

4. Toolset

- **Postman:** For functional API testing and automated collections.
- **Cypress:** For end-to-end browser automation and smoke suites.

- *Apache JMeter*: For establishing a performance baseline [186, Conversation History].
- *Git/GitHub*: For version control and pull request management of all QA artifacts

5. Manual Test Execution Report (20 Cases)

This table provides the complete execution results for all 20 manual scenarios.

TC-ID	Test Scenario	Technique	Expected Result	Actual Result	Status
TC_01	Successful Owner Login	EP	User redirected to main dashboard.	Redirected as expected.	Pass
TC_02	Invalid Password (EP)	EP	"Invalid credentials" error message.	Error message appeared.	Pass
TC_03	Empty Field Validation	Logic	Alert for both blank fields.	Only email alert shown.	Pass*
TC_04	Session Persistence	Functional	Remain logged in after tab refresh.	User remained logged in.	Pass
TC_05	Past Appointment Booking	Logic	System should block past times.	Appointment confirmed.	FAIL
TC_06	Multi-Branch Visibility	Functional	Show data for every owned branch.	All branches visible.	Pass
TC_07	Branch Switching UI	UI	UI updates for selected branch.	Dashboard updated.	Pass
TC_08	Unauthorized Branch Access	Security	Deny access/redirect to authorized.	Redirected correctly.	Pass
TC_09	GST Billing Logic (18%)	Functional	Correct CGST/SGST 18% split.	Calculations accurate.	Pass
TC_10	Insufficient Points	Functional	Block redemption if balance low.	Transaction blocked.	Pass
TC_11	Exact Balance Redemption	Functional	Redemption accepted; balance becomes 0.	Balance moved to zero.	Pass
TC_12	Zero Point Earn Rule	Logic	No points added for 0 total.	No points added.	Pass
TC_13	PDF Invoice	UI	PDF matches	Layout/data	Pass

	Consistency		dashboard data.	consistent.
TC_14	Sequential Numbering	Logic	Sequential financial-year numbers.	Numbering is sequential. Pass
TC_15	State Transition (Refund)	Transition	Move to "Refunded"; points deducted.	State changed correctly. Pass
TC_16	Loyalty Member Search	Functional	Correct profile shown in search.	Member found by name. Pass
TC_17	Mobile Responsiveness	UI	Elements stack for screens <768px.	Layout stacks correctly. Pass
TC_18	Staff Role List View	UI	List shows all staff and roles.	List displayed correctly. Pass
TC_19	Navigation Integrity	Functional	"Explore" link loads without 404.	Page loaded correctly. Pass
TC_20	Analytics Graph Loading	UI	Graphs load without delay.	Visibility was immediate. Pass

*Note: TC_03 was marked Pass despite the observation that an alert for the password field was missing; a UI suggestion was noted.

6. Final Bug Report Pack (5 Reports)

Based on the failures and exploratory findings above, the following defects were documented:

1. **BUG-001 (Medium/Logic):** System allows booking appointments for past times (TC_05).
2. **BUG-002 (Low/Privacy):** Branch filter does not isolate loyalty members; shows all customers globally.
3. **BUG-003 (Low/BVA):** Password validation fails to enforce the 33rd character limit.
4. **BUG-004 (High/Functional):** Missing "Delete" functionality for appointments, preventing data cleanup.
5. **BUG-005 (Medium/Async):** Refunded invoices show persistent "PDF is still generating" error for Credit Notes.

7. Postman (API Testing)

- **Contract Validation:** Verified successful **200 OK** responses for core identity services (Cognito) with a response time of **44ms**.
- **Integration Risk:** Identified a failure in the **DELETE /appointments/cancel** request, suggesting the backend does not handle specific cancellation states correctly.

8. Cypress (E2E Automation)

Your Cypress suite follows the architecture of running in the same **run loop** as the application, providing native access to DOM objects.

- **Key Phase:** Each test covers: **Set up state** (`cy.visit()`), **Take action** (`cy.get().type()`), and **Assertion** (`cy.should()`).
- **Advanced Features:** Used **"Automatic Waiting"** to eliminate flakiness and **"Time Travel"** to replay test steps.

9. Architecture & Serverless Risk Assessment

- **Code Boundaries:** Separating the **Lambda handler** (outer adapter) from the **Business Logic** (inner functions) ensures easier debugging of async failures.
- **Async/Integration Risks:** The **"sad path"** for the Klodr workflow (e.g., webhook failure) requires verification of **retry logic** and traceability through CloudWatch logs.

10. Performance Baseline

- **Tool:** Apache JMeter.
- **Key Result:** Establishing a **20ms average response time** with **0.00% error rate** under a load of 5 virtual users.

11. QA Summary Report

- **Strategic Summary:** Testing successfully transitioned from manual black-box techniques (EP, BVA, Transitions) to automated regression safety.
- **Risk Assessment:**

- **Async Risk:** The Credit Note PDF generation error (BUG-005) indicates a failure in background processing.
- **Integration Risk:** The "Cancel Appointment" API failure shows a contract gap between the UI and backend.
- **Git Workflow:** Artifacts are managed via a professional Git workflow, including **Feature Branching**, meaningful **Commits**, and **Pull Requests** for mentor review.

Final Conclusion

- **Conclusion:** The module is stable for core tasks but requires immediate fixes for **BUG-001 (Logic)** and **BUG-004 (Data Integrity)** before release.